I chose the healthcare industry. The healthcare industry is where human health is improved, healed, and checked on. Hospitals, medical devices, insurance providers, and more are all within the healthcare industry. The purpose is disease prevention, emergency and long-term care providence, medication delivery, therapy (physical, occupation, mental), and other health technologies. Healthcare is meant to be a service to everybody. This includes but not limited too: individuals, families, entire populations and civilizations, animals too. Healthcare has been relevant for so long and it will continue being so because the elderly need medical services, chronic diseases rise globally, pandemics can spring up out of nowhere so society needs to respond quickly when that happens, etc. In short, healthcare is essential because it directly affects quality of life, life expectancy, economic stability, and societal well-being.

Healthcare must serve everyone, and biased AI can worsen existing inequalities instead of improving outcomes. AI systems in healthcare learn from historical medical data but that data often contains racial, gender, socioeconomic, and geographic biases. This creates several ethical risks:

- AI models may perform worse on underrepresented groups (e.g., darker-skinned patients in dermatology datasets).
- If past medical decisions were biased, AI may reinforce those patterns.
- Communities already facing disparities may receive lower-quality AI-driven care.

Health information is deeply personal. Misuse or exposure can harm patients socially, financially, and emotionally. Healthcare AI relies on enormous amounts of personal, sensitive medical data, electronic health records, imaging, genetic information, wearable-device data, and more. Key ethical concerns include:

- Medical data is highly valuable and vulnerable.
- Patients may not know how their data is being used to train AI systems.
- Even “anonymized” data can sometimes be traced back to individuals.

According to your powerpoint Mr. Ganti, AI in healthcare can be like this:

- Integration of multi-omics data for holistic health understanding
    - We’ll get personalized medicine and early diagnosis of diseases for example, but we’ll need to counter data-integration complexity and ethical and privacy concerns because of sensitive genetic info.
- AI in regenerative medicine and tissue engineering
    - 3D bioprinting is enhanced and stem-cell research is accelerated, but you’ll need to prepare for limited data quality of AI models and ethical issues surrounding human-cell manipulation for example
- Quantum computing and AI for complex biological simulations
    - Molecular and genomic simulations can be performed and we can get real-time analysis of massive biological datasets. However, quantum computing is going to be cost and energy demanding and transparency issues can be brought to light because of how the algorithms will make decisions.

- Brain-computer interfaces and neuroprosthetics
    - Rehabilitation and assistive robotics improved through AI-sync control but invasive procedures can pose long-term threats to health and surgery

I was most surprised about brain-computer interfaces and neuroprosthetics. I think the AI advancements in healthcare are very exciting and I can't wait to see what will happen next. While it would be very exciting, I don't know how I would be able to adapt to the changes, especially not really having very little medical knowledge unlike my father (He's a nurse practitioner). Even so, I'm not one to run away from a challenge so I would work in this industry and find ways to further push the industry with AI advancements.

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